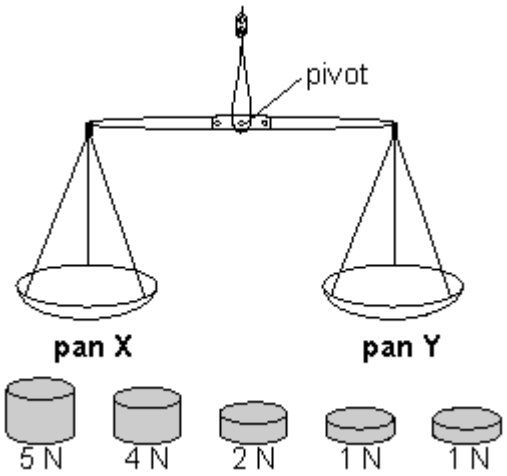
Q1. Ellie has a set of scales and some weights as shown below.



Ellie puts two weights in pan X and one weight in pan Y. The scales balance.

(a) Which weights could be in pans X and Y?

pan X: ..... and .....

pan Y: .....

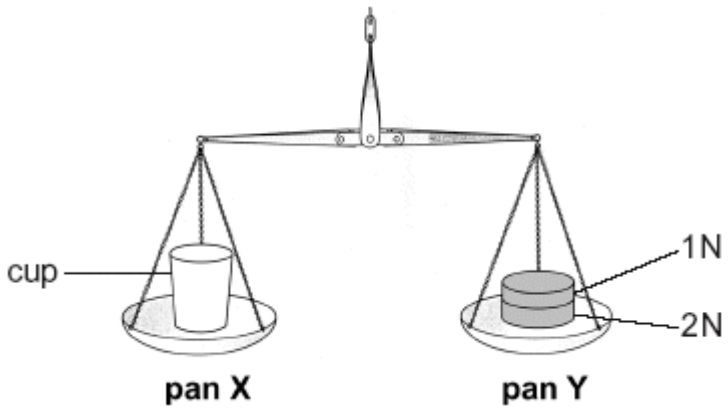
1 mark

(b) Ellie removes all the weights from the scales.
She then puts a cup on pan X.
In which direction will pan Y move?

.....

1 mark

(c) She puts weights into pan Y so the scales balance.

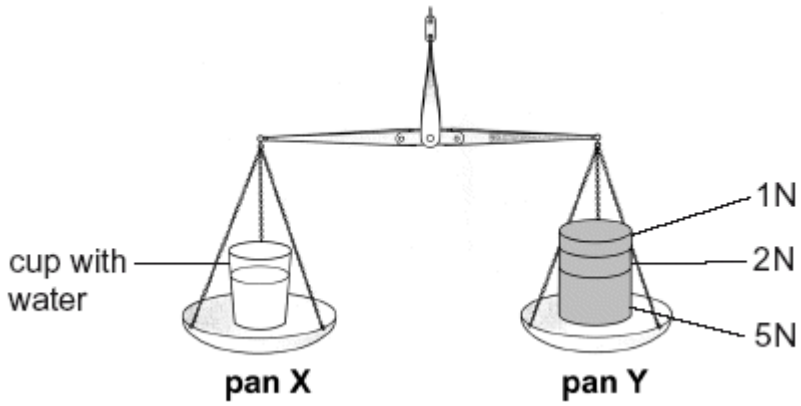


How much does the cup weigh?

..... N

1 mark

(d) Ellie puts some water in the cup.
She then adds some more weights to pan Y to make the scales balance.



(i) How much do the cup **and** water weigh?

..... N

1 mark

(ii) How much does the water weigh?

..... N

1 mark
maximum 5 marks

Q1.

(a) **either**

- *pan X:* 1N and 1N
  pan Y: 2N

or

- *pan X:* 4N and 1N
  pan Y: 5N

1 (L3)

three *weights are required for the mark*
units are not required for the mark
the weights in pan X can be in either order

(b) • up

accept '↑'
'X will go down' is insufficient

1 (L3)

(c) • 3 *N*

1 (L3)

(d) (i) • 8 *N*

1 (L3)

(ii) • 5 *N*

accept the answer to (di) minus the answer to (c)
accept '8-3'

1 (L4)

[5]